

ASSIGNMENT #02

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
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Due Date: Friday, February 13th TA's Mailbox

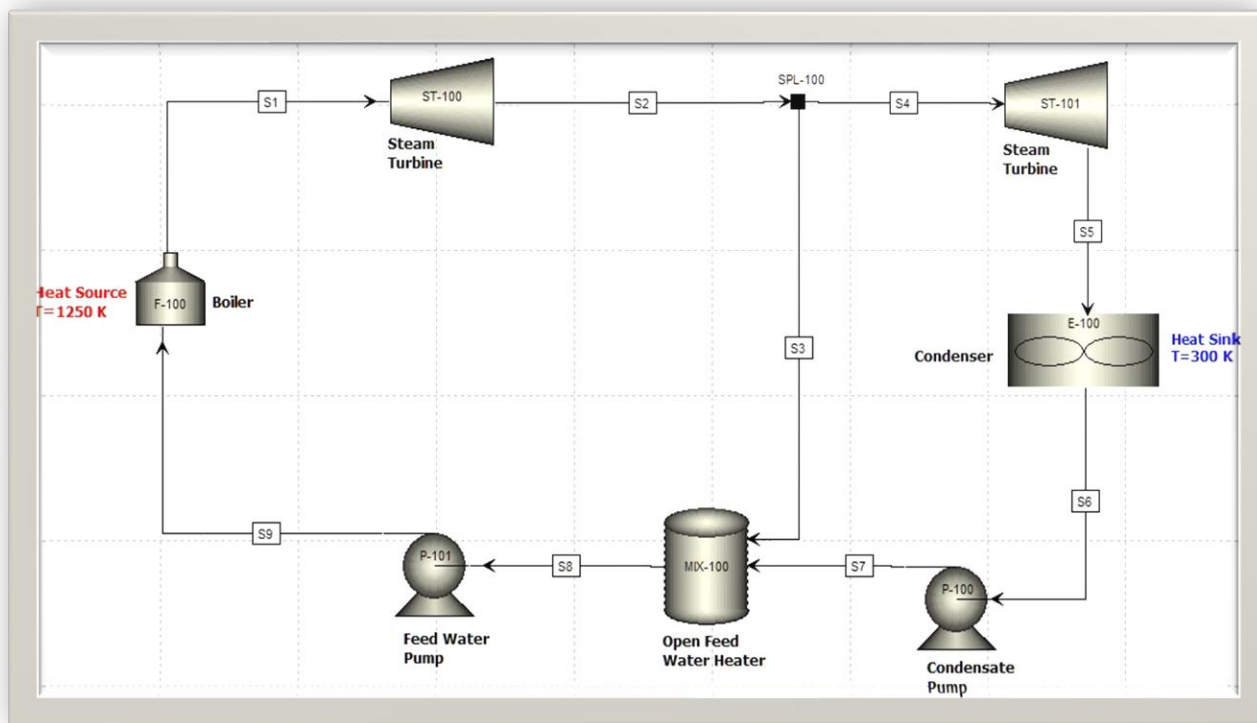
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: The Regenerative Rankine Cycle for Power Production.

The Athena Power Consulting Group proposed a unique design for the O'Brien Hospital in Athens OHIO, to provide **500kW** electric power. The design is shown in the process flowsheet diagram below:



The boiler(F-100) generates steam at 40 bar and 450 °C. The superheated steam expands in the High Pressure (**HP**) steam turbine (ST-100) down to 2 bar. The discharge from the second Low Pressure (**LP**) steam turbine (ST-101) is condensed in the vacuum condenser (E-100). The pressure drop in the condenser is negligible and the effluent is saturated liquid at 40 °C. A fraction of the HP turbine (ST-100) discharge steam at 200 kPa is extracted from the turbine and mixed with the discharge of the condensate pump (P-100) in the Open Feedwater Heater (**OFWH**). as shown in the diagram, and the resulting subcooled liquid mixture is then pumped to 4 MPa by the feedwater pump (P-101) before being supplied to the boiler (F-100). As a young engineer your purpose is to evaluate this new design and compare its performance to the standard Rankine Cycle design. Prepare a detailed report of your solution and calculate the following:

1. The steam flow rate in MMSCFD through the cycle required to provide the turbine output power of 500kW.
2. The mass fraction of the bled steam required to provide 1 °C subcooled liquid condition at the outlet of the Open Feed Water Heater (**OFWH**).
3. The overall thermal efficiency of this power plant.
4. The lost work in kW in every unit along with the total.

5. The thermodynamic efficiency of this power plant.
6. The maximum theoretical efficiency of this thermal cycle.
7. The cost of utilities (electricity cooling water and natural gas) using the cost data below.
8. Perform an optimization to minimize the annual cost of utilities. This optimization requires that you find the HP turbine discharge pressure that gives the lowest cost of the plant utilities. Consider a pressure range between 1 and 10 bar for the optimization.
9. The emissions of CO₂ in gigatons (Gt) per operating year.
10. Find through research the CO₂ emissions per year worldwide in gigatons. Make sure you report the year for which you have found this information. What percentage is the contribution of this power plant to the worldwide emissions.
11. Calculate the Natural gas required flow in MMSCFD
12. Determine the air requirement in MMSCFD
13. Determine the NO_x emissions in kg per operating year

Miscellaneous Data:

1. Turbines (ST-100, ST-101) efficiency: 0.85
2. Pumps (P-100, P-101) efficiency: 0.75
3. Cooling water available at 20 °C
4. Temperature rise in the cooling water: 10 °C
5. Boiler (F-100) uses natural gas as fuel with high heating value(HHV) of 55.5 MJ/kg.
6. Composition of natural gas: 100% Methane
7. For the methane combustion, the air entering the burner is essentially ambient air with temperature 25 °C and pressure 1 atm
8. Air Composition: 21 mol% O₂ and 79 mol% N₂
9. Oxygen concentration in the flue gas: 3 mol% on a dry basis
10. Assume an NO_x emission factor of 0.05 lb NO_x/MMBtu (HHV)
11. Boiler Efficiency is 85%
12. Operating year 8,000 hours
13. Cost of electricity: 0.0775 \$/kWh
14. Cost of cooling water: 0.354 \$/GJ
15. Cost of Natural gas: 4.25 \$/GJ

Enter your results in the Table similar to the one below:

Steam flow rate, MMSCFD	2.0921
Mass fraction of the bled steam (stream S3)	0.1295
Thermal efficiency, %	33.74%
Thermodynamic Efficiency, %	45.095%
Lost work, kW	608.76kW
Maximum theoretical efficiency, %	76%
Minimum Annual Cost of utilities, \$/year	199.9 million \$/year

You may wish to have more entries in the table, that you consider important

Minimum Report Requirements:

1. Prepare detailed analysis of the high- and low-pressure turbines
2. Prepare detailed analysis of the Boiling Feed Water pump
3. Prepare detailed analysis of the Open Feed Water Heater
4. Prepare a table with the list of all equipment and the lost work associated with each one
5. Prepare a full material and energy balance using the ENG Calculations worksheet

CHE 301-Assignment 2

Daohn Kim

February 2026

Title: Thermodynamic Analysis of a Regenerative Rankine Cycle

Abstract

The report will analyze the components in the given cycle and calculate the power generated and the costs in the operation of the cycle. Each component has different properties and roles, so let us know the relationship of how the components affect each other. The evaluation for the cycle will be compared to the standard Rankine cycle, and it is engaged to find out the way or conditions to achieve the lowest costs in terms of economy.

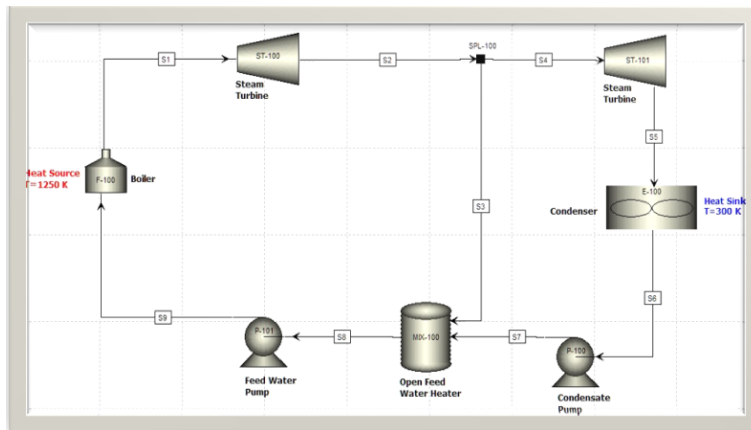


Figure 1: The designed flow diagram for the O'Bleness Hospital

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1 Introduction

1.1 Background

This problem is introduced with the regenerative cycle from the Athena Power Consulting Group, as shown in Figure 1 above. The given cycle consists of one HP turbine and one LP turbine, and two kinds of pumps (condensate pump and feed water pump) with a condenser and boiler. In addition, there is one more steam which is extracted from the HP turbine and is connected to the open feed water heater, so a total of 9 streams are flowing. The values of T_c and T_h are 300K and 1250K.

To solve a bunch of the problems, the assumption of conditions in the cycle is steady-state, and the pump and turbine are adiabatic processes.

As in following the conditions, we can use the common equations:

Enthalpy balance equation:

$$-\Delta \dot{m}H + \dot{Q} + \dot{W}_s = 0 \quad (1)$$

Entropy balance equation:

$$-\Delta \dot{m}S + \sum_i \frac{\dot{Q}_i}{T_i} + \dot{S}_{gen} = 0 \quad (2)$$

1.2 HP& LP turbines roles

The High-Pressure Steam Turbine gets the high power, superheated steam directly from the boiler. The steam keeps maintaining the superheated vapor phase almost throughout the turbine. On the other hand, the low-pressure steam turbines is connected to the condenser, so the expansion of the steam is planned to drop the pressure. That pressure will be lower than 1 atm, which is called sub-atmospheric pressure, and it is determined by the temperature of the available cooling water. In short, the hp turbine is fitted into the boiler, and the LP turbine is for the condenser.

Using Google AI

To obtain detailed information about the turbine system and its properties, I searched Google AI. It analysed the HP turbine to extract high-temperature superheated steam, which will be used to preheat the fluid. For the LP turbine, as the temperature drops toward the saturation point, the steam will be in the saturated phase, and due to the increased specific volume of steam at low pressures, the LP turbine features larger blade diameters compared to the HP turbine. In Calculation, the fact that the pressure of exhaust steam from the LP turbine should be determined by the water saturation point because the steam is saturated state.

1.3 The boiling feed water pump

I searched the company 'PumpWorks', which is making the boiling feed water pump. They say that the pump supplies water to the boiler to be converted to steam. The boiling feed water pump will work to draw water from the feed water tank and increase its pressure before sending it to the boiler, and maintain a consistent flow rate under high-pressure conditions to ensure stable steam production.

1.4 The open feed water heater

Using Google AI

The only difference is that handling two inlets in this cycle. I looked up the role of the open feed water using Google AI. An open feed water pump is a centrifugal pump used to transport water from a deaerator or open feedwater heater—where it is mixed with exhaust steam—to a boiler, operating at high pressures and temperatures.

2 Steam flow rate to output turbine power of 500kW

2.1 Need of basis of the mass fraction of S3

Before we solve the problem of the steam flow rate value, we need to first know the mass fraction of S3.

The output power of the turbine is equal to the summation of the work rates by two turbines, which are calculated with enthalpy equations, as using equation (1), and the Q is negligible because the turbine has proceeded in adiabatic:

$$Power = W_s = \Delta \dot{m} H$$

The power is determined by mass and molar enthalpy by temperature and pressure, which means that mass plays an important role. When I see the steam 4 of inlet steam of the second turbine, the mass flow rate (or molar flow rate) will be changed differently to the previous flow rate in the first turbine. Therefore, we need to know the mass flow rate of S4 and the mass fraction of S3 will help us determine what the mass flow rate of S4 is. That's why I am going to solve the problems of the mass flow rate and the mass fraction together.

The trace of equations in Excel

In Excel, to watch the clearly visible data of the relation between them, the function of "trace the formula" is used, and the figure 2,3 shows the web which indicates the values affect each other. The shown trace represents that the output power of the turbines is descending to the molar mass, and the molar fraction(mass fraction) connects to the enthalpy balance. To set those two equation getting correct values, the related values can be determined.

	A	B	C	D	E	F	G	H	I	J	K	L
	ENG Calculations	Ideal Gas										
</												

Figure 2: The trace from the output power of turbine equation

ENG Calculations	Ideal Gas										
	S1	S2 (ideal)	S2	S3	S4	S5 (ideal)	S5	S6	S7 (ideal)	S7	S8
Stream Properties											
Molar Flow, kmol/s	0.0289	0.0289	0.0289	0.0037	0.0252	0.0252	0.0252	0.0252	0.0252	0.0252	0.0289
Mass Flow, kg/s											
Volume Flow, m³/s											
Temperature, C	450.0000		134.8362	134.8362	134.8362			40.0000	40.0047	40.0199	119.2308
Pressure, bar	40.0000	2.0000	2.0000	2.0000	2.0000	0.0738	0.0738	0.0738	2.0000	2.0000	2.0000
Vapor Fraction	1.0000	0.9664	1.0000	1.0000	1.0000	0.8628	0.8935	0.0000			0.0000
Molecular Weight, kg/kmol											
Molar Enthalpy, kJ/kmol	60011.902	47421.451	49307.638	49307.638	49307.638	40427.258	41759.315	3016.658	3020.086	3021.226	9015.728
Vapor Phase		48753.821				46377.193	46377.193				
Liquid Phase		9092.186				3016.658	3016.658				
Molar Entropy, kJ/(kmol K)	125.003	125.003	129.772	129.772	129.772	129.772	134.026	10.307	10.307	10.310	27.370
Vapor Phase		128.390				148.773	148.773				
Liquid Phase		27.564				10.307	10.307				
Molar Density, kmol/m³											
Enthalpy Flow, kJ/s	1736.96	1372.54	1427.14	184.83	1242.31	1018.57	1052.13	76.00	76.09	76.12	260.95
Entropy Flow, kJ/(s K)	3.62	3.62	3.76	0.49	3.27	3.27	3.38	0.26	0.26	0.26	0.79
Availability, kJ/s	651.55		300.32	38.89	261.42		39.09	-1.90		-1.81	23.29
Duty [Power], kJ/s [kW]		-364.41	-309.82			-223.74	-190.18	-976.12	0.09	0.12	
Efficiency			0.85				0.85			0.75	
Compressibility Factor			Ws				Ws2	Qc		Ws3	
User Data and Calculations	Assumed			Lw		Efficiency		Cost			
Entropy balance in ST-100	0.0000	mass frac	0.1295	Lost Work ST-100 Turbine, MW		41.4136	thermal	33.7421	fuel	4.2500	
Entropy balance in ST-101	0.0000	discharge P	2.0000	Lost Work ST-101, MW		32.1520	thermodynam	45.0954	cooling	0.3540	976.1249
Output power in turbine, kw	500.0000	mass flow	2.0921	Lost Work Condensor, MW		40.9900	max theoretic	76.0000	electric	0.0775	
Entropy balance in P-100	0.0000			Lost Work P-100		0.0259					
Entropy balance in two inlets and outlet	0.0000			Lost Work P-101		0.5277					
Entropy balance in P-101	0.0000			Lost Work Boiler		493.6522					
				Tot Lost Work		608.7613					

Figure 3: The trace from the Enthalpy balance in two inlets and one outlet

2.2 Process of S1 to S8

S1 to S2

The first step is getting the entropy of S1 and S2(ideal). For S1, the temperature and pressure are 450 °C, 40bar, and the pressure of ideal S2 is 2bar with an unknown temperature. The x quality of ideal S2 will be

Stream Properties	S1	S2 (ideal)	S2	S3	S4	S5 (ideal)	S5	S6	S7 (ideal)	S7	S8
Molar Flow, kmol/s	0.0289	0.0289	0.0289	0.0037	0.0252	0.0252	0.0252	0.0252	0.0252	0.0252	0.0289
Mass Flow, kg/s											
Volume Flow, m ³ /s											
Temperature, C	450.0000		134.8362	134.8362	134.8362			40.0000	40.0047	40.0199	119.2308
Pressure, bar	40.0000	2.0000	2.0000	2.0000	2.0000	0.0738	0.0738	0.0738	2.0000	2.0000	2.0000
Vapor Fraction	1.0000	0.9664	1.0000	1.0000	1.0000	0.8628	0.8935	0.0000			0.0000
Molecular Weight, kg/kmol											
Molar Enthalpy, kJ/kmol	60011.902	47421.451	49307.638	49307.638	49307.638	40427.258	41759.315	3016.658	3020.086	3021.226	9015.728
Vapor Phase		48753.821				46377.193	46377.193				
Liquid Phase		9092.186				3016.658	3016.658				
Molar Entropy, kJ/(kmol K)	125.003	125.003	129.772	129.772	129.772	129.772	134.026	10.307	10.307	10.310	27.370
Vapor Phase		128.390				148.773	148.773				
Liquid Phase		27.564				10.307	10.307				
Molar Density, kmol/m ³											
Enthalpy Flow, kJ/s	1736.96	1372.54	1427.14	184.83	1242.31	1018.57	1052.13	76.00	76.09	76.12	260.95
Entropy Flow, kJ/(s K)	3.62	3.62	3.76	0.49	3.27	3.27	3.38	0.26	0.26	0.26	0.79
Availability, kJ/s	651.55		300.32	38.89	261.42		39.09	-1.90		-1.81	23.29
Duty [Power], kJ/s [kW]		-364.41	-309.82			-223.74	-190.18	-976.12	0.09	0.12	
Efficiency			0.85				0.85			0.75	
			Ws				Ws2	Qc		Ws3	
Compressibility Factor											

Figure 4: The process S1 to S8

the target value to be changed by completing the equations.

$$\begin{aligned}
S_{S1} &= S_{\text{steam}}(450^\circ\text{C}, 40 \text{ bar},) = 125.003 \text{ kJ/kmol K} \\
S_{S2,v} &= S_{\text{steam}}(, 2 \text{ bar}, \text{vapor}) = 128.390 \text{ kJ/kmol K} \\
S_{S2,l} &= S_{\text{steam}}(, 2 \text{ bar}, \text{liquid}) = 27.564 \text{ kJ/kmol K} \\
x * S_{S2,v} + (1 - x) * S_{S2,l} &= 125.003 \text{ kJ/kmol K (lever rule)}
\end{aligned}$$

The Q and S_{gen} are zero because the ideal S2 indicates the situation of a reversible case. The relation between S1 and ideal S2 is by equation (2)

$$\begin{aligned}
-\Delta \dot{m} S &= 0 \\
S_1 - S_2 &= 0
\end{aligned}$$

The entropy balance of S1 and ideal S2 is set to zero, and the x quality will be the changed value. Then, the x results in 0.9664 by using Excel Goal Seek. Next step, the data is given that the exhaust steam from the LP turbine is saturated liquid at 40 °C. The efficiency of ST-100 is 0.85, so when we used the efficiency equation,

$$\mu = \frac{\dot{W}_s}{\dot{W}_{rev}} = 0.85 \quad (3)$$

In order to get the Ws, the values of the enthalpy should be obtained.

$$H_{S1} = S_{\text{steam}}(450^\circ\text{C}, 40 \text{ bar},) = 60011.902 \text{ kJ/kmol}$$

- ideal S2

The temperature and quality is not known here, so I tried to figure out the solution with x quality as assumed that the steam will turn into the mixture, but the result is the superheated vaport at 2 bar. I can start to

Stream Properties	S1	S2 (ideal)	S2
Molar Flow, kmol/s	0.0289	0.0289	0.0289
Mass Flow, kg/s			
Volume Flow, m³/s			
Temperature, C	450.0000		134.8362
Pressure, bar	40.0000	2.0000	2.0000
Vapor Fraction	1.0000	0.9664	1.0000
Molecular Weight, kg/kmol			
Molar Enthalpy, kJ/kmol	60011.902	47421.451	49307.638
Vapor Phase		48753.821	
Liquid Phase		9092.186	
Molar Entropy, kJ/(kmol K)	125.003	125.003	129.772
Vapor Phase		128.390	
Liquid Phase		27.564	
Molar Density, kmol/m³			
Enthalpy Flow, kJ/s	1736.96	1372.54	1427.14
Entropy Flow, kJ/(s K)	3.62	3.62	3.76
Availability, kJ/s	651.55		300.32
Duty (Power), kJ/s [kW]		-364.41	-309.82
Efficiency			0.85
			Ws
Compressibility Factor			
User Data and Calculations	Assumed		
Entropy balance in ST-100	0.0000	mass frac	0.1295
Entropy balance in ST-101	0.0000	discharge P in	2.0000
Output power in turbine, kw	500.0000	mass flow	2.0921
	0.0000		
Entropy balance in P-100	0.0000		
Entropy balance in two inlets and outlet	0.0000		
Entropy balance in P-101	0.0000		

Figure 5: The process from S1 to S2

show the equation with the unknown temperature. the

$$H_{S2,v} = H_{\text{steam}}(, 2 \text{ bar}, \text{vapor}) = 48753.821 \text{ kJ/kmol}$$

$$H_{S2,l} = H_{\text{steam}}(, 2 \text{ bar}, \text{liquid}) = 9092.186 \text{ kJ/kmol}$$

$$x * S_{S2,v} + (1 - x) * S_{S2,l} = 47421.451 \text{ kJ/kmol K (lever rule)}$$

- S2

The temperature and quality is not known here, so I tried to figure out the solution with x quality as assumed that the steam will turn into the mixture, but the result is the superheated vaport at 2 bar. I can start to show the equation with the unknown temperature.

$$H_{S2} = H_{\text{steam}}(T_2, 2 \text{ bar},)$$

$$\dot{W}_s = H_{S2} - H_{S1} \dot{W}_{rev} = H_{S2,rev} - H_{S1} \mu = \frac{\dot{W}_s}{\dot{W}_{rev}} = \frac{H_{S2} - H_{S1}}{H_{S2,rev} - H_{S1}}$$

By using the Goal Seek, the temperature of S2 will be changed to make efficiency of 0.85. The result of T is 134.8362 °C.

S4 to S5

$$P_6 = \text{Water}P_{\text{sat}}(40^\circ\text{C}) = 0.0738\text{bar}$$

The pressure drop between S5 and S6 is negligible, so the pressure of S5 is the same as the pressure of S6, 0.0738 bar. In the S2, the steam is split into two steams but there is no energy change like work, and heat. The value of pressure and temperature from the HP steam turbine will be maintained in the two streams of S3 and S4, but the molar flow rate will be unknown. Let me talk about the molar flow rate later, so we can keep moving to the next with a random number of molar flow rates. Now, we know the S4 and S5 conditions at all, and the efficiency of ST-101 is 0.85. As I showed the above in the step of S1 to S2, if I repeat the same process to get the quality of S5 instead of the temperature in the balance equations due to the mixture in S5, the result of x is 0.8935.

S6 to S8

The pressure of S7 and S8 should be 2bar because it is mentioned in the problem that the discharge steam of the hp turbine at 2bar is mixed with the discharge steam of the condenser. The constant pressure will be displayed on the pressure line, and the S7 will be sub-cooled liquid from the condenser. The temperature is set as the changed value.

$$\begin{aligned} S_{S6} &= S_{\text{steam}}(40, 0.0738 \text{ bar},) \\ S_{S7,rev} &= S_{\text{steam}}(T_7, 2 \text{ bar},) \\ S_{S6} - S_{S7,rev} &= 0(\text{reversible}) \end{aligned}$$

The temperature of S7 in the reversible is 40.0047 °C.

$$\begin{aligned} H_{S6} &= H_{\text{steam}}(40, 0.0738 \text{ bar},) \\ H_{S7,rev} &= H_{\text{steam}}(40.0047, 2 \text{ bar},) \\ H_{S7} &= H_{\text{steam}}(T_2, 2 \text{ bar},) \\ \dot{W}_s &= H_{S7} - H_{S6} \\ \dot{W}_{rev} &= H_{S7,rev} - H_{S6} \\ \eta &= \frac{\dot{W}_{rev}}{\dot{W}_s} \end{aligned} \tag{4}$$

The efficiency equation for pumps is defined oppositely. Therefore, the efficiency of P-100,0.75 is calculated by the reciprocity.

$$\eta = \frac{\dot{W}_{rev}}{\dot{W}_s} = \frac{H_{S7,rev} - H_{S6}}{H_{S7} - H_{S6}} = 0.75$$

As it is said that steam is required to provide a 1 °C subcooled liquid condition at the outlet of the Open Feed Water Heater in the problem, the temperature of S8 will be defined as follows:

$$T_8 =$$

Calculation

- Assumption: mass fraction of S3, mass flow rate
- Calculation tool: using the Excel Goal Seek(Che calculator)

The results even look like the correct answer, but the goal power is 500kW, and we still have those unclear variables of mass fraction and mass flow rate. In the figure 3, there are two inlets and one outlet without

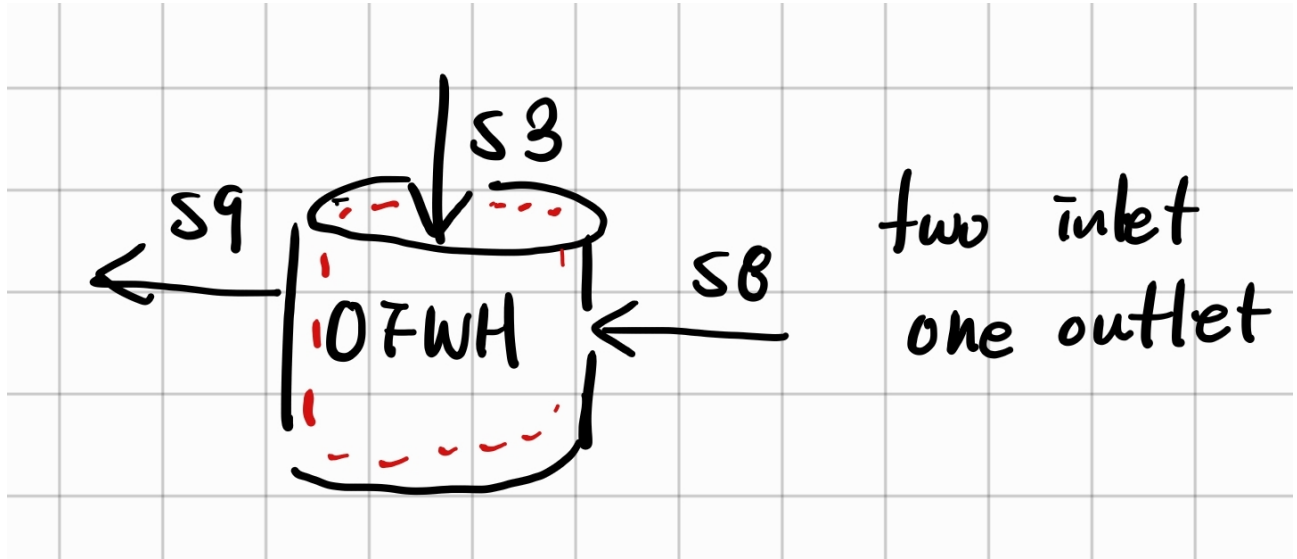


Figure 6: The flow steams in the OFWH

any W and Q . The enthalpy equation for the OFWH is

$$-\Delta \dot{m}H = 0 = H_3 + H_7 - H_8 T_8 = WaterTsat(2bar) = 119.2308CH_{S8} = H_{steam}(T_8, 2bar,)$$

The values of H_3 , H_7 , and H_8 are calculated, and the enthalpy is multiplied by the molar flow rate in each state. The mass fraction is the changed value, and the enthalpy balance in the figure 3 is set to zero. The fraction results in 0.1295. The second equation, which we can use, is the equation for the output turbine power of 500kW.

$$Power = W_s + W_{s2} = -500kW$$

By using the Goal Seek again, the 500kw will be set as the value to the power variable, by changing the mass flow rate. The Goal Seek showed the mass rate is 0.0289 kmol/s. If it is converted into MMSCFD, it will be 2.0921 MMSCFD.

3 Efficiency

3.1 Process of S8 to S1

In the process of the boiling feed water pump, the water steam is directed right to the boiler, so the pressure of S9 should be planned to fit in the S1 boiler. The pressure of S9 and S1 is 40 bar, and the phases for the two streams are vapor. The pump efficiency should be dealt with in the same way as the first pump is calculated to get the temperature of the exhaust steam of the pump. If we ignore the repeated calculations above and check the value obtained directly, the temperature is 119.8610 °C.

S8	S9 (ideal)	S9	S1
0.0289	0.0289	0.0289	0.0289
119.2308	119.5448	119.8610	450.0000
2.0000	40.0000	40.0000	40.0000
0.0000		1.0000	1.0000
9015.728	9087.976	9112.097	60011.902
27.370	27.369	27.431	125.003
260.95	263.04	263.74	1736.96
0.79	0.79	0.79	3.62
23.29		25.55	651.55
	2.09	2.79	1473.22
		0.75	
		Ws4	Qh

Figure 7: The process of S8 to S1

3.2 Calculation of thermal efficiency

$$\eta_{th} = \frac{W_{net}}{Q_H} \quad (5)$$

The net work is the summation of the work by two turbines and two pumps. The turbines have negative values of work because it works by the system. The works which called as W_s, W_{s2}, W_{s3} , and W_{s4} as written in Excel, will be described as

$$W_{net} = W_s + W_{s1} + W_{s2} + W_{s3} = (-309.82kW - 190.18kW + 0.12kW + 2.79kW) = -490.09kW$$

$$\eta_{th} = \frac{W_{net}}{Q_H} = \frac{490.09kW}{1473.22kW} = 33.74$$

3.3 Lost work

$$LW = -\Delta\dot{m}(H - T_0S) + \sum \dot{W}_s + \dot{Q} \left(1 - \frac{T_0}{T}\right) \quad (6)$$

Unit	Duty [kW]
W_s (HP)	-309.82
W_{s2} (LP)	-190.18
W_{s3} (P-100)	0.12
W_{s4} (P-101)	2.79
Q_c	-976.12
Q_h	1473.22

Figure 8: The process of S8 to S1

$$\begin{aligned}
LW_{turbine} &= -\Delta\dot{m}(B) + \sum \dot{W}_s[availability : B] \\
LW_{condenser} &= -\Delta\dot{m}(B) \\
LW_{pump} &= -\Delta\dot{m}(B) + \sum \dot{W}_s \\
LW_{boiler} &= -\Delta\dot{m}(B) + \dot{Q}(1 - T_0/T_H)
\end{aligned}$$

The lost work formula is formed in each component. Especially in the condenser, $T_0 = T_c$, that's why the calculation of Q is cancelled out and only leaves the change of the availability.

Lw	
Lost Work ST-100 Turbine, MW	41.4136
Lost Work ST-101, MW	32.1520
Lost Work Condensor, MW	40.9900
Lost Work P-100	0.0259
Lost Work P-101	0.5277
Lost Work Boiler	493.6522
Tot Lost Work	608.7613

Figure 9: Lost Work in each component, kW

3.4 Thermodynamic efficiency

This problem's goal is a negative number, $GW \leq 0$

$$\eta_{thermo} = \frac{GW}{GW - LW} * 100 \quad (7)$$

Efficiency	
thermal	33.7421
thermodynam	45.0954
max theoretic	76.0000

Figure 10: The efficiency in the cycle

Figure 6 organizes each lost work and the total lost work, which will be used as LW in the above equation.

$$\eta_{thermo} = \frac{GW}{GW - LW} * 100 = \frac{-500kW}{-500kW - 608.7613kW} * 100 = 45.09$$

The thermodynamic efficiency is used to identify if the cycle is working well, so the percentage of 45.09 means that lost work is created in the cycle. However, when I search up the average thermodynamic efficiency, it says "Thermodynamic efficiency in a cycle, defined as the ratio of useful work output to heat input, averages between 30 and 50." The percentage of 45.09 is higher than 30 percent, so it could be evaluated as a good working cycle.

3.5 Maximum theoretical efficiency

The maximum theoretical efficiency is applied when it is reversible, so the thermal efficiency cannot go over the maximum theoretical efficiency. The formula for the maximum efficiency is:

$$\eta = (1 - \frac{T_c}{T_H}) * 100 \quad (8)$$

The T_c and T_H are mentioned from 1. Introduction part, which are 300K and 1250K, respectively.

$$\eta = (1 - \frac{T_c}{T_H}) * 100 = (1 - \frac{300K}{1250K}) * 100 = 76.0$$

4 Costs in the cycle

4.1 Analysis of costs

For costs, there are three kinds of costs considered in the problem: electricity, fuel, and cooling water costs. The fuel costs occur in the boiler to increase the temperature of the stream. Constant combustion of the fuel might be needed to maintain a constant T_H in the boiler, where the steam takes the Q heat. In the condenser, the cost will be charged to cool the water. Lastly, the electricity costs are related to the pumps, which need the energy to raise the steam.

4.2 Calculation for costs

$$\text{Cost}_{\text{fuel}} = Q_{\text{fuel}} \times (4.25 \$/\text{GJ}) = Q_H / 0.85 \times (4.25 \$/\text{GJ}) \quad (9)$$

For the cost of fuel, the heat of fuel is defined by dividing by efficiency with the Q_h , which means the process of heat exchange between the boiler and steam has the lost work, and the lost work is cancelled out from the heat of fuel to know the amount of the heat transferred to the steam.

$$\text{Cost}_{\text{cooling}} = Q_c \times (0.3540 \text{ \$/GJ}) \quad (10)$$

$$\text{Cost}_{\text{elec}} = W_{\text{pump}}/\eta \times (4.25 \text{ \$/kWh}) \quad (11)$$

The work from the pump is not how much work is consumed to make the pump work, which is adjusted by the efficiency. The value of work before calculation with the efficiency of the pump will be a correct work value to be used in the cost of electricity. The table of the costs will be described like below:

	Cost	\$/GJ	GW	\$/s	annual cost
	fuel	4.2500	1.7332	7.3661	212143759.5482
	cooling	0.3540	0.9761	0.3455	9951788.5904
	electric	21.5278	0.0039	0.0834	2401884.6638

Figure 11: The table of the costs: fuel, cooling, and electricity

Result

The result of the total annual costs is 224497432.8025.

4.3 Optimization to get the lowest cost

To minimize the annual cost of utilities, an optimization was performed by varying the High-Pressure (HP) turbine discharge pressure within the range of 1 to 10 bar. The change of the HP turbine discharge pressure will affect to most of the pressures in the cycle. The colored cells for pressure represent the changed pressure

S1	S2 (ideal)	S2	S3	S4	S5 (ideal)	S5	S6	S7 (ideal)	S7	S8	S9 (ideal)	S9	S1
0.0287	0.0287	0.0287	0.0034	0.0253	0.0253	0.0253	0.0253	0.0253	0.0253	0.0287	0.0287	0.0287	0.0289
450.0000		117.5760	117.5760	117.5760			40.0000	40.0047	40.0171	98.6315	112.5770	112.8815	450.0000
40.0000	1.0000	1.0000	1.0000	1.0000	0.0738	0.0738	0.0738	1.0000	1.0000	1.0000	40.0000	40.0000	40.0000

Figure 12: The influence of the discharge pressure change

dependent on the discharge pressure.

4.3.1 Cost Objective Function

The total annual cost (C_{total}) is defined as the sum of the costs for natural gas, cooling water, and net electricity over 8,000 operating hours:

$$C_{\text{total}} = 8000 \times (\dot{m}_{\text{fuel}} \cdot C_{\text{fuel}} + \dot{Q}_{\text{out}} \cdot C_{\text{CW}} + \dot{W}_{\text{net}} \cdot C_{\text{elec}}) \quad (12)$$

4.3.2 Optimization Parameters

The following fixed constraints and efficiencies were applied during the sensitivity analysis:

- **Turbine Efficiency:** 0.85
- **Output Power of Turbine:** 500kW
- **Pump Efficiency:** 0.75
- **Boiler Efficiency:** 85%
- **Fuel:** 100% Methane (HHV = 55.5 MJ/kg)

To watch the change of the process by the discharge pressure, the first try will be with the pressure below 2bar. When I tried to change the discharge pressure to 1.6 bar and organized all data to follow the previous process, as the pressure was at 2bar,

S1	S2 (ideal)	S2	S3	S4	S5 (ideal)	S5	S6	S7 (ideal)	S7	S8	S9 (ideal)	S9	S1
0.0287	0.0287	0.0287	0.0034	0.0253	0.0253	0.0253	0.0253	0.0253	0.0253	0.0287	0.0287	0.0287	0.0289
450.0000		117.5760	117.5760	117.5760			40.0000	40.0047	40.0171	112.3199	112.5770	112.8815	450.0000
40.0000	1.6000	1.6000	1.6000	1.6000	0.0738	0.0738	0.0738	1.6000	1.6000	1.6000	40.0000	40.0000	40.0000
1.0000	0.9543	1.0000	1.0000	1.0000	0.8655	0.8935	0.0000			0.0000		1.0000	1.0000
60011.902	46742.669	48732.483	48732.483	48732.483	40545.348	41759.315	3016.658	3019.448	3020.376	8487.629	8557.149	8580.319	60011.902
	48572.853				46377.193	46377.193							
	8563.886				3016.658	3016.658							
125.003	125.003	130.149	130.149	130.149	130.149	134.026	10.307	10.307	10.310	26.014	26.005	26.065	125.003
	129.739				148.773	148.773							
	26.212				10.307	10.307							
1722.64	1341.75	1398.86	167.31	1231.56	1024.65	1055.33	76.24	76.31	76.33	243.64	245.63	246.30	1736.96
3.59	3.59	3.74	0.45	3.29	3.29	3.39	0.26	0.26	0.26	0.75	0.75	0.75	3.62
646.18		278.08	33.26	244.83		39.21	-1.91		-1.83	19.62		21.84	651.55
	-380.89	-323.78			-206.90	-176.22	-979.10	0.07	0.09		2.00	2.66	1490.66
		0.85			0.85				0.75			0.75	
		Ws				Ws2	Qc		Ws3			Ws4	Qh
											</		

Figure 13: The entire process at the discharge pressure of 1.6 bar

The total annual costs are over 226,000,000 dollars, which is bigger than the costs at 2bar. At 10 bar of the discharge pressure, the entire process will be changed like:

The phase of the exhaust steam from the HP turbine is superheated vapor. Therefore, we need to change the way to get the enthalpy and entropy values.

$$S_{S2,rev} = S_{steam}(T_2, 10bar)$$

$$H_{S2,rev} = H_{steam}(T_2, 10bar)$$

The entropy balance between reversible S2 and S1 is zero.

$$S_{S1} = S_{S2,rev}$$

The temperature of reversible S2 reached 252.89 °C, in the state of superheated vapor. The efficiency of the HP turbine is fixed to 0.85 as before, so the efficiency cells are set to 0.85, resulting in the temperature of S2, 278.96 °C, by using the Excel Goal Seek. As adjusting all data to follow the fixed conditions, the mass fraction of S3 is 0.2085. The total annual costs are around 199,000,000 dollars. That cost is the smallest among what I tried. The change of the discharge pressure between 1 bar and 10 bar leads to the change of

	S1	S2 (ideal)	S2	S3	S4	S5 (ideal)	S5	S6	S7 (ideal)	S7	S8	S9 (ideal)	S9	S1
3	0.0296	0.0296	0.0296	0.0046	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0296	0.0296	0.0296	0.0289
4														
5														
6	450.0000	139.0436	180.3943	180.3943	180.3943			40.0000	40.0047	40.0305	137.8730	138.1002	138.3629	450.0000
7	40.0000	3.5000	3.5000	3.5000	3.5000	0.0738	0.0738	0.0738	3.5000	3.5000	3.5000	40.0000	40.0000	40.0000
8	1.0000		1.0000	1.0000	1.0000	0.8555	0.8927	0.0000			0.0000		1.0000	1.0000
9														
0														
1	60011.902	49217.077	50833.364	50833.364	50833.364	40111.622	41725.076	3016.658	3022.477	3024.414	10448.499	10509.156	10529.370	60011.902
2		49210.213				46377.193	46377.193						50448.097	
3		10525.626				3016.658	3016.658						19589.546	
4	125.003	125.028	128.764	128.764	128.764	128.764	133.917	10.307	10.306	10.312	30.930	30.905	30.954	125.003
5		125.009				148.773	148.773							
6		31.117				10.307	10.307							
7														
8														
9	1778.40	1458.51	1506.40	233.92	1272.48	1004.09	1044.48	75.51	75.66	75.71	309.63	311.43	312.03	1736.96
0	3.70	3.71	3.82	0.59	3.22	3.22	3.35	0.26	0.26	0.26	0.92	0.92	0.92	3.62
1	667.10		361.66	56.16	305.50		38.80	-1.89		-1.73	34.66		36.84	651.55
2		-319.90	-272.00			-268.39	-228.00	-968.96	0.15	0.19		1.80	2.40	1424.93
3			0.85				0.85			0.75			0.75	
4			Ws				Ws2	Qc		Ws3			Ws4	Qh
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4.3.3 Results

To analyse the change of the annual cost, I think that the mass fraction increases as the discharge pressure in the HP turbine increases. The mass fraction indicates the amount of the hot steam, which is mentioned in the Introduction. The hot steam from the HP turbine helps to reduce the energy needed to heat it up in the boiler, so the increase in the amount of hot steam means a help for reducing the work in the boiler and pump. In the given range of the optimization pressure, 10 bar is the best answer for the pressure to bring the lowest cost.

Analysis by formula

When the discharge pressure of the HP turbine ($P_{discharge}$) increases, the temperature and enthalpy of the extracted steam entering the open feedwater heater (h_3) increase accordingly.

Applying the energy balance to the open feedwater heater:

$$yh_3 + (1 - y)h_2 = h_4 \quad (13)$$

where:

- y = mass fraction of extracted steam
- h_2 = enthalpy of feedwater before mixing
- h_3 = enthalpy of extracted steam
- h_4 = enthalpy after mixing (boiler inlet enthalpy)

If h_3 increases, either:

- the required extraction fraction y decreases for a fixed h_4 , or
- h_4 increases for a fixed y .

In both cases, the required boiler heat input decreases:

$$Q_{in} = \dot{m}(h_{boiler \ exit} - h_4) \quad (14)$$

Therefore, regeneration reduces the boiler heat duty directly.

Estimation with the dominant variable

The dominant operating cost in utility systems is the natural gas cost (C_{NG}).

The total fuel cost can be expressed as:

$$\text{Total Cost} \approx \dot{m}_{fuel} C_{NG}$$

Since fuel mass flow rate is related to boiler heat duty:

$$\dot{m}_{fuel} = \frac{\dot{Q}_{in}}{\eta_{boiler} \cdot HHV}$$

Thus,

$$\text{Total Cost} = \frac{\dot{Q}_{in}}{\eta_{boiler} \cdot HHV} C_{NG} \quad (15)$$

As the extraction pressure approaches approximately 10 bar, the feedwater heater performance is maximized, leading to minimized $\dot{Q}_{in}$. Because fuel cost is the dominant term in the cost function, reducing $\dot{Q}_{in}$ significantly lowers the total operating cost.

As the extraction pressure increases:

- Pump work (W_{pump}) decreases due to a reduced pressure difference.
- LP turbine work ($W_{turbine}$) may decrease slightly because less expansion occurs.

However, the fuel-saving benefit from reduced Q_{in} is mathematically greater than the power loss penalty from reduced turbine work. Therefore, from an economic optimization perspective, an extraction pressure near 10 bar provides the optimal operating condition.

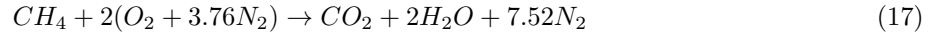
5 Fuel

5.1 Emission of CO2

The emission of CO2 is directly explained by the combustion reaction of methane. The combustion of natural gas, assumed to be 100% methane (CH_4), follows the stoichiometric reaction with oxygen:



To determine the actual air requirement, we account for the nitrogen (N_2) present in the ambient air (21% O_2 and 79% N_2 by mole). The complete reaction with air is expressed as:



The reaction between methane and oxygen gas occurs and produces CO2 and H2O, so for this calculation, the N_2 gas doesn't need to be handled. Therefore, we can see that equation (16) instead of (17) simply states the molar ratio of methane and CO2 is 1:1. That indicates the methane molar flow rate should be the same as the CO2 molar flow rate. In this step, there is no information for the molar flow rate, so after the question of the natural gas required flow is solved, we can get it. To easy flow calculation, I can draw the mass flow rate of methane of 0.0312kg/s.

$$\begin{aligned} \dot{n}_{CO_2} &= \frac{\dot{m}_{CH_4}}{M_{CH_4}} \times (M_{CO_2}) = \frac{0.0312kg/s}{16.08kg/kmol} \times (44.009kg/kmol) = 0.0854kg/s \\ \dot{n}_{CO_2} * 8000hr * \frac{60min}{1hr} \frac{60s}{1min} &= 2459249.2kg \\ &= 2459249.2kg \times \frac{10^{-12}Gt}{1kg} = 2.46 * 10^{-6}Gt \end{aligned}$$

5.2 Global CO₂ Contribution Analysis

To evaluate the environmental impact of the proposed 500kW power cycle, the amount of CO2 emissions in the global scale should help to give the basis information and understand the macro-view for CO2 emissions, and the above part showed the math results for the CO2 emission in one cycle. This part will look at how the CO2 emissions are reported globally.

Based on the latest report from the "International Energy Agency (IEA)", they estimated that the CO2 emission in 2023 grew at 37.4 Gigatonnes and antipated the trend of increasing emissions will be in the crisis.

This increase was primarily driven by extreme drought events affecting hydropower and a continued reliance on fossil fuels in emerging economies.

Contribution: Percentage Calculation The annual CO_2 emission from this specific cycle was calculated in Question 9 as $2.46 \times 10^{-6} \text{ Gt}$. The relative contribution of this plant to the total global emissions is determined using the following

$$\text{Global Contribution (\%)} = \left(\frac{\text{Annual Plant Emission (Gt)}}{37.4 \text{ Gt}} \right) \times 100 \quad (18)$$

$$\text{Global Contribution (\%)} = \left(\frac{2.46 \times 10^{-6} \text{ Gt}}{37.4 \text{ Gt}} \right) \times 100 = 6.6 \times 10^{-6}$$

The result of contribution of 6.6×10^{-6} means a small part contributed to the global CO_2 emission, and the CO_2 emission from this cycle can be negligible as an aspect of the global scope.

5.3 Natural gas required flow

- **Boiler Efficiency:** 85%
- **Fuel:** 100% Methane (HHV = 55.5 MJ/kg)

The efficiency of the boiler is 0.85. When there is a heat exchange between the heat generated by the fuel and transferred to the steam, the relation between the heat of the fuel and the steam is defined with the efficiency:

$$\dot{Q}_{fuel} = \dot{Q}_{steam} / 0.85 \quad (19)$$

$$\begin{aligned} \dot{Q}_{fuel} &= \dot{Q}_{steam} / 0.85 = 1733.2 \text{ kW} \\ \dot{m}_{fuel} &= \dot{Q}_{steam} / HHV = \frac{1733.2 \text{ kW}}{55.5 \text{ MJ/kg}} = 0.0312 \text{ kg/s} \end{aligned}$$

To convert kg/s to MMSCFD unit,

$$\frac{\dot{m}_{fuel}}{M_{fuel}} * 86400 \text{ s/day} * 379.48 \text{ scf/kmol} * \frac{1}{10^6} = 7809.04 \text{ MMSCFD}$$

5.4 Air requirements flow

As per the system requirements, the burner is operated with excess air to maintain an oxygen concentration of 3 mol% in the flue gas on a dry basis. This necessitates an air-fuel ratio higher than the stoichiometric ratio of 9.52:1 (moles of air per mole of fuel).

$$\dot{n}_{fuel} = \frac{0.04148 \text{ kg/s}}{16.04 \times 10^{-3} \text{ kg/mol}} = 2.586 \text{ mol/s} \quad (20)$$

To satisfy the condition of 3 mol% O_2 in the dry flue gas, the required molar air-to-fuel ratio is:

$$AF_{molar} = 11.45 \quad (21)$$

gas flow rate, MMSCFD	6906.3606
NOX emission	0.1187
per operating year	949.5989

Figure 19: NOx emission

(This is greater than the theoretical stoichiometric ratio of 9.52.)

$$\dot{n}_{air} = \dot{n}_{fuel} \times AF_{molar} = 2.586 \text{ mol/s} \times 11.45 = 29.61 \text{ mol/s} = 29.61 \times 10^{-3} \text{ kmol/s} \quad (22)$$

The conversion to MMSCFD: 2.27MMSCFD

5.5 Nox emissions

An NOx emission factor is given in the problem so by converting the unit to match each unit of the data, the equation is the emission factor times the gas flow rate of methane.

$$\dot{m}_{NO_x} = EF_{NO_x} \times \dot{m}_{CH_4} \quad (23)$$

where:

- EF_{NO_x} = NO_x emission factor
- $\dot{m}_{CH_4}$ = methane mass flow rate

If annual emission is required:

$$m_{NO_x, annual} = EF_{NO_x} \times \dot{m}_{CH_4} \times t_{operation} \quad (24)$$

For 8000 operating hours per year:

$$m_{NO_x, annual} = EF_{NO_x} \times \dot{m}_{CH_4} \times (8000 \times 3600) \quad (25)$$

The NOx emission per year is 949.59 kg.

6 Conclusion

6.1 Answers for the assignment

The answers through a lot of calculations can be organized in the table (1).

6.2 Evaluation to the given cycle & Comparison with the standard Rankine cycle

The proposed regenerative cycle demonstrates a clear thermodynamic advantage over the standard Rankine cycle by increasing feedwater temperature prior to boiler entry, thereby reducing the required heat input and improving overall fuel utilization. With a thermal efficiency of 33.74, the cycle outperforms a simple Rankine configuration operating under similar temperature limits due to regeneration and optimized extraction

Table 1: Summary of Final Calculated Results

questions	answers
Mass fraction (y)	0.1295
Total mass flow rate	0.0289 kmol/s
Mass flow rate (fuel)	0.0312 kg/s
Fuel molar flow rate	2.586 mol/s
Required air molar flow rate	29.61 mol/s
Air flow rate	2.27 MMSCFD
Thermal efficiency (η_{th})	33.74 %
Thermodynamic efficiency	45.09 %
Maximum theoretical efficiency	76.0 %
Total minimum annual cost	\$224,497,432.80
Annual CO_2 emission	2.46×10^{-6} Gt
Global CO_2 contribution	6.6×10^{-6} %
Annual NO_x emission	949.59 kg
Optimal HP discharge pressure	10 bar

pressure. Economically, the dominant reduction in boiler heat duty leads to lower fuel consumption and minimized annual operating cost at an HP discharge pressure of 10 bar. However, the additional components (open feedwater heater and extraction control) increase system complexity compared to the conventional Rankine cycle. In addition, the environmental concerns cannot be mentioned through the results of the low contribution percentage.

References

International Energy Agency. (2024). CO2 emissions in 2023: Emissions grew in 2023, but clean energy is limiting the growth. <https://www.iea.org/reports/co2-emissions-in-2023/emissions-grew-in-2023-but-clean-energy-is-limiting-the-growth>

PumpWorks, “What is a Boiler Feed Pump How Does it Work?” — <https://www.pumpworks.com/what-is-a-boiler-feed-pump-how-does-it-work/>